

CORE DATA STRUCTURES

Arrays & Linked Lists

Sequential storage and connected data structures in
computer science



Structural Foundations

Two fundamental approaches to organizing data in memory



Arrays

Contiguous memory blocks with direct index-based random access.

$O(1)$ Access Time



Linked Lists

Nodes scattered in memory, connected sequentially via pointers.

$O(1)$ Insertion / Deletion

Anatomy of an Array

Fixed-size collection stored in contiguous memory addresses



✓ Instant index lookup

✗ Costly resizing & shifts

Anatomy of a Linked List

Dynamic node chain connected via memory pointers



✓ Dynamic size allocation

✗ $O(n)$ sequential traversal

Complexity Trade-offs

Algorithmic efficiency across common operations

Operation	Array	Linked List
Access (Index)	O(1)	O(n)
Insertion (Beginning)	O(n)	O(1)
Deletion (Beginning)	O(n)	O(1)
Memory Overhead	Low	Higher (Pointers)

Choosing the Right Structure

Use arrays for fast lookups and static datasets. Use linked lists for frequent insertions and dynamic memory needs.

Questions & Discussion